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NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Introduction To Machine Learning - IITKGP
(course)



Course outline

About
NPTEL ()

How does an
NPTEL
online
course
work? ()

Week 0 ()

Week 1 ()

Week 2 ()

Week 3 ()

Week 4 ()

Week 5 ()

Week 6 ()

Week 7 ()

☐ Lecture 36 :
Introduction to
Computational
Learning

Week 7 : Assignment 7

The due date for submitting this assignment has passed.

Due on 2024-09-11, 23:59 IST.

Assignment submitted on 2024-09-11, 22:08 IST

1)

2 points

Which of the following options is / are correct regarding the benefits of ensemble model?

1. Better performance
2. More generalized model
3. Better interpretability

- A) 1 and 3
B) 2 and 3
C) 1 and 2
D) 1, 2 and 3

- ☐ A)
☐ B)
☒ C)
☐ D)

Yes, the answer is correct.

Score: 2

Accepted Answers:

C)

2)

2 points

Theory (unit?
unit=59&lesso
n=60)

○ Lecture 37 :

Sample
Complexity :
Finite
Hypothesis
Space (unit?
unit=59&lesso
n=61)

○ Lecture 38: VC

Dimension
(unit?
unit=59&lesso
n=62)

○ Lecture 39:

Introduction to
Ensembles
(unit?
unit=59&lesso
n=63)

○ Lecture 40:

Bagging and
Boosting (unit?
unit=59&lesso
n=64)

○ Tutorial 7

(unit?
unit=59&lesso
n=65)

○ Week 7 :

Lecture
Material (unit?
unit=59&lesso
n=66)

● **Quiz: Week 7 : Assignment 7 (assessment? name=148)**

○ Feedback
Form for Week
7 (unit?
unit=59&lesso
n=134)

● Assignment 7
Solution (unit?
unit=59&lesso
n=158)

In AdaBoost, we give more weights to points having been misclassified in previous iterations. Now, if we introduce a limit or cap on the weight that any point can take (for example, say we introduce a restriction that prevents any point's weight from exceeding a value of 10). Which among the following would be the effect of such a modification?

- A) It will have no effect on the performance of the Adaboost method.
- B) It makes the final classifier robust to outliers.
- C) It may result in lower overall performance.
- D) None of these.

☐ A)

☒ B)

☒ C)

☐ D)

Yes, the answer is correct.

Score: 2

Accepted Answers:

B)

C)

3)

2 points

Identify whether the following statement is true or false:

"Boosting is easy to parallelize whereas bagging is inherently a sequential process."

A) True

B) False

☐ A)

☒ B)

Yes, the answer is correct.

Score: 2

Accepted Answers:

B)

4)

2 points

Considering the AdaBoost algorithm, which among the following statements is true?

- A) In each stage, we try to train a classifier which makes accurate predictions on a subset of the data points where the subset contains more of the data points which were misclassified in earlier stages.
- B) The weight assigned to an individual classifier depends upon the weighted sum error of misclassified points for that classifier.
- C) Both option A and B are true
- D) None of them are true

☐ A)

☐ B)

☒ C)

☐ D)

Yes, the answer is correct.

Score: 2

Accepted Answers:

C)

Week 8 ()

Assignment
Solution ()Download
Videos ()Problem
Solving
Session -
July 2024 ()

5)

2 points

Which of the following is FALSE about bagging?

- A) Bagging increases the variance of the classifier
- B) Bagging can help make robust classifiers from unstable classifiers.
- C) Majority Voting is one way of combining outputs from various classifiers which are being bagged.

☒ A)☐ B)☐ C)

Yes, the answer is correct.

Score: 2

Accepted Answers:

A)

6)

2 points

Suppose the VC dimension of a hypothesis space is 6. Which of the following are true?

- A) At least one set of 6 points can be shattered by the hypothesis space.
- B) Two sets of 6 points can be shattered by the hypothesis space.
- C) All sets of 6 points can be shattered by the hypothesis space.
- D) No set of 7 points can be shattered by the hypothesis space.

☒ A)☐ B)☐ C)☒ D)

Yes, the answer is correct.

Score: 2

Accepted Answers:

A)

D)

7)

2 points

Identify whether the following statement is true or false:

“Ensembles will yield bad results when there is a significant diversity among the models.”

A) True

B) False

☐ A)☒ B)

Yes, the answer is correct.

Score: 2

Accepted Answers:

B)

8)

2 points

Which of the following algorithms is not an ensemble learning algorithm?

- A) Random Forest
- B) Adaboost
- C) Decision Trees

- ☐ A)
- ☐ B)
- ☒ C)

Yes, the answer is correct.

Score: 2

Accepted Answers:

C)

9)

2 points

Suppose you have run Adaboost on a training set for three boosting iterations. The results are classifiers h_1 , h_2 , and h_3 , with coefficients $\alpha_1 = 0.2$, $\alpha_2 = -0.3$, and $\alpha_3 = -0.2$. For a given test input x , you find that the classifiers results are $h_1(x) = 1$, $h_2(x) = 1$, and $h_3(x) = -1$. What is the class returned by the Adaboost ensemble classifier H on test example x ?

- A) 1
- B) -1

- ☒ A)
- ☐ B)

Yes, the answer is correct.

Score: 2

Accepted Answers:

A)

10)

2 points

Generally, an ensemble method works better, if the individual base models have _____? (Note: Individual models have accuracy greater than 50%)

- A) Less correlation among predictions
- B) High correlation among predictions
- C) Correlation does not have an impact on the ensemble output
- D) None of the above.

- ☒ A)
- ☐ B)
- ☐ C)
- ☐ D)

Yes, the answer is correct.

Score: 2

Accepted Answers:

A)